proof $B^n_v(n-x) = (-1)^v B^n_v(x)$

Primero se demuestra que: $\varphi(n;\tau;n-x) = \varphi(n;-\tau;x)$

$$\varphi(n;\tau;n-x) = \frac{\tau^n e^{\tau(n-x)}}{(e^\tau-1)^n} = \frac{\tau^n e^{-\tau x}\cdot e^{\tau n}}{(e^\tau-1)^n}\cdot\frac{(-1)^n}{(-1)^n}\,\frac{e^{\tau n}}{e^{\tau n}}$$

$$\varphi(n;\tau;n-x) = \frac{(-\tau)^n e^{-\tau x}\,\cancel{e^{\tau n}}}{(1-e^\tau)^n}\,\frac{e^{\tau n}}{\cancel{e^{\tau n}}} = \frac{(-\tau)^n e^{-\tau x}}{\left[e^{-\tau}(1-e^\tau)\right]^n}$$

$$\boxed{\varphi(n;\tau;n-x) = \frac{(-\tau)^n e^{-\tau x}}{(e^{-\tau}-1)^n} = \varphi(n;-\tau;x)}$$

Luego si:

$$\varphi(n;\tau;x) = \sum_{v=0}^{\infty}\frac{\tau^v}{v!}B^n_v(x)$$

$$\varphi(n;\tau;n-x) = \sum_{v=0}^{\infty}\frac{\tau^v}{v!}B^n_v(n-x) = \sum_{v=0}^{\infty}\frac{(-\tau)^v}{v!}B^n_v(x)$$

Es evidente: $\boxed{B^n_v(n-x) = (-1)^v B^n_v(x)}$